

In all, between November 1942 and Paulus' surrender at the end of January 1943, the Luftwaffe lost 490 machines, including 266 Ju 52s and 165 He 111s.

The Eastern Front

From 1943 onward, the Luftwaffe suffered an ultimately unsustainable rate of attrition on the eastern front and during the Allied strategic bombing offensive over Germany. Because German aircraft production belatedly expanded under the direction of new Nazi industrial chief Albert Speer, German airmen were able to continue inflicting heavy losses on their enemies, but it was a losing battle against mounting odds.

The Germans on the whole sent their less experienced pilots and more obsolescent aircraft to the Eastern Front, reserving the latest and best for the defense of Germany. Yet Luftwaffe pilots in Russia recorded astonishing kill rates: top ace Erich Hartmann claimed 352 victories, and six other German pilots are credited with over 200 each. However, Soviet fighters were increasingly a match for their enemy – the Yakovlev Yak-9 could hold its own against Focke-Wulfs and Messerschmitts and denied the Luftwaffe superiority over the battlefield.

“Tank-busting” aircraft on both sides – the Soviet Il-2 Shturmovik and Lavochkin La-5, and the Luftwaffe's Henschel Hs 129s and Ju 87 Stukas equipped with armor-piercing cannon – played a major part in the great tank battle of Kursk in July 1943 and in the other armored clashes that followed as the Soviets rolled the German armies back. A single Stuka pilot, Hans-Ulrich Rudel, was estimated to have destroyed more than 500 Soviet tanks. But the German panzers also suffered grievously under air attack, and the Soviets were better able to replace lost armor. It was the same story in the struggle for the air. In all, the Luftwaffe claimed to have shot down 44,000 Soviet aircraft, yet German airplanes were still outnumbered on the Eastern Front throughout the later stages of the war.

The D-Day landings

In the west, air power was central to the success of the D-Day invasion of Normandy in June 1944 and the Allied drive across Europe that followed. At the start of the war the American and British air forces were ill-prepared for providing direct support to ground forces. Even if they accepted the goal of achieving air superiority over the battlefield, they were little interested in providing air support. Senior air commanders, who resented the idea of being at the service of the army, had done nothing to develop techniques for

Yakovlev Yak-3

“Avoid combat below 5,000m [16,000ft] with Yakovlev fighters lacking an oil cooler under the nose.”

LUFTWAFFE DIRECTIVE, EASTERN FRONT

SOVIET DESIGNER ALEKSANDR YAKOVLEV's Yak-3 fighter earned its reputation as a supreme dogfighter in the desperate large-scale battles on the Eastern Front in 1944–45. To its shock and dismay, the Luftwaffe found that, at least at lower altitudes, it was superior to both the Focke-Wulf Fw 190 and the Messerschmitt Bf 109. On one occasion in July 1944, a squadron of 18 Yak-3s that encountered a force of 30 Luftwaffe fighters is claimed to have succeeded in shooting down 15 of them for the loss of only one of its own aircraft.

Small coolers in wing roots replace single oil cooler under nose

Thick coat of hard-wearing wax polish

Coolant radiator duct

Pneumatic wheels

Soviet red star marking

Retractable tailwheel

All-metal monocoque fuselage structure

COMPACT PLANE

The Soviet designer Aleksandr Yakovlev, created a formidable fighter by cutting down the Yak-1's airframe and making the wings smaller.

LIGHTWEIGHT DOGFIGHTER

The Yak-3 was so sleek and lightweight that it achieved high performance without a powerful engine. Both armor and armament were sacrificed to keep the weight down. This aircraft was flown by Free French airmen operating with the Soviets.